Educational robot Kiddo learns to draw to enhance interactive handwriting scenario for primary school children.

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Abstract—Handwriting and drawing skills play a significant role in early childhood education. Recent studies suggest that robots currently being used in education are social robots where most of them don't write directly as the way humans do by holding a pen and moving it over a paper, besides, use an indirect medium such as touch screen as the shared environment which creates a noticeable barrier in natural interaction. In this work, we develop a new robot named Kiddo that sits on a desk, has an animatronic face, can write with a whiteboard marker, and provides a shared environment where the robot and a child can write simultaneously. The robot learns to write shapes by maneuvering its hand just like we humans do which creates a vivid image to the child interacting with it. We have used CNN based model to recognize and draw 100 shapes collected from the National Curriculum Bengali, English, and Mathematics textbook of Bangladesh. To learn new shapes from children's demonstration an artificial intelligence system was build which can generate path trajectories from a camera image. We experimented with three kids where they drew 18 known and 4 new shapes on the shared whiteboard and Kiddo was also able to recognize and draw 17 of the known shapes and learn to draw all the new shapes from the children's demonstration in both simulation and in real.

Index Terms—Educational Robot, Human Robot Interaction, Robot Writing, Artificial Intelligence

I. Introduction

Children start to learn writing and drawing since their commencement of education. In the first few years, they learn writing basic shapes for example alphabet, math symbols, line, curve, etc. For ages, demonstrating drawing skills by holding a pen and moving the hand over a paper makes an authentic image to the child to help developing motor mimicry and cognitive development. This scenario is known widely and researchers have identified several areas where using a social robot as a teacher or a care receiver can enhance children's learning activity significantly [1] [2]. However, using the popular social robots posses a large gap in natural interaction especially in handwriting skills because most of the robots don't write directly in a shared environment instead they use another medium such as touch screens for writing and taking input. Writing on a touch-screen can cause children to lose interest in holding the pen which is essential for good handwriting and drawing skills.

In this study, we are providing a solution by developing a new robot named kiddo which sits on a desk in a conventional table-chair environment and provides a shared environment where children and the robot can write simultaneously. Figure 1 shows a scenario where Children, as well as Kiddo writes on the shared whiteboard.

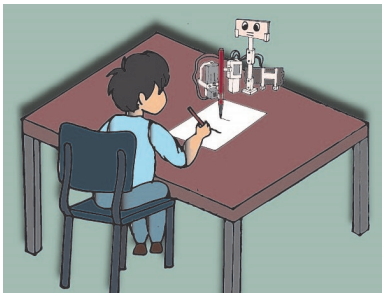

Figure 1. The robot Kiddo sits on a table and a child sitting in a chair and both writes on a shared whiteboard.

Our main contributions are, firstly, a new educational robot Kiddo, which has a robotic arm and provides a whiteboard where children and the robot can write in a shared environment. Kiddo has an animating face and can communicate verbally. Secondly, we have identified one hundred basic shapes from elementary, grade 1 and grade 2 textbooks of NCTB (National Curriculum and Textbook Board, Bangladesh) and used deep learning to make the robot recognize those shapes of six categories (Bengali vowel, Bengali consonant, Bengali digit, English digit, English alphabet, and basic geometric shapes). We also created a dataset so that Kiddo can write those shapes.

Thirdly, we have developed a noble approach to find segmented drawing trajectories from an image so that Kiddo can learn drawing new shapes from children's demonstration. As children are keen to give commands [3] Kiddo's learning ability from children might trigger protégé effect to the child [4].

The structure of the paper is as follows. First, we discuss related works in educational robots (Section II) and then described Kiddo (Section III) and then our method to make Kiddo recognize and draw known and unknown shapes from children demonstration (Section IV). Experiment setup (Section V) and the outcome of the experiments (Section VI) are discussed in (section VII).

II. RELATED WORK

How a social robot can be used to enhance children's learning activity is an active research area. Researchers identified different focus area and most of the researchers used robots mainly as a teacher and as a poor student which create an opportunity for the students to teach it to induce protégé effect. In a review paper Belpaeme, Tony, et al. showed that in education research most-used robots are Nao, Keepon, Wakamaru, iCat, Robovie, etc and all of the robots have humanoid features such as head, eye, mouth, arms or legs [5]. Hood et al. (2015) devolved a robotic learning agent using NAO robot which employs the learning by teaching paradigm where children teach handwriting to stimulate meta-cognition, empathy, and increased self-esteem where they invest more effort into learning [6] [7]. Another robotic learning agent has previously been developed by Tanaka and Matsuzoe [8] where children learn vocabulary by teaching the NAO robot to act out verbs. The robot is tele-operated and mimics the actions that the children teach it. El-Barkouky, Ahmed, et al has developed a system where they used light polarization property for detection of the computer screen and provided an environment where children and robot write side by side [9]. Potkonjak, Veljko developed a system to mimic human-like fatigue behavior while writing with a humanoid robot [10]. Sojib,Noushad shows a simulation robot that can draw a few shapes [11].

III. THE ROBOT: KIDDO

The robot Kiddo consists of a robotic arm, an android cell-phone, a standard paper clipboard as the base, a whiteboard, a whiteboard marker, a camera, a controller circuit and a power supply. A computer can be connected with Kiddo over a USB cable to run intelligent applications. The robotic arm has 2 Degree of Freedom (DOF) and an end-effector of 1 DOF which is responsible for holding the writing marker and lifting it up or down. We have developed an app which shows an animatronic face on the phone display and all the voice recognition and robot speaking is done through this app. The size of the whiteboard is 21cm x 22.50cm and divided equally into two part one for the robot drawing and one for the children interacting with it. Figure 2 shows the real robot (left) and the simulated robot (right). Kiddo also provides a button interface with two buttons to get manual feedback from the users. The arm consists of two dynamixel AX-12A and one SG90 servo motors. A Arduino Nano works as the Dynamixel motor driver and a Arduino mega works as the main controller to drive all the hardware which connects to a laptop with a USB. The robot requires 12V power supply. The entire Kiddo fits in a box of 35cm x 23cm x 23 cm (LxWxH).

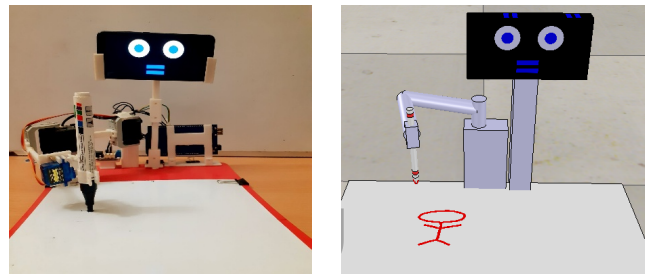

Figure 2. Educational robot Kiddo has an animatronic face and share a whiteboard with children. The real robot (left) and the simulated Kiddo(right) in VREP works in the same way.

A. Simulated robot and the real robot

Kiddo was first simulated in VREP [12]. The simulated version and the real robot works in a similar way. The simulated robot has a workspace of 300x200 pixels where the workspace in the real robot is 120x110mm. Linear coordination is used to fit the simulated workspace into the real one.

B. Robot space and Cartesian space

Link lengths, $L_1 = 103mm, L_2 = 105mm$
Joint variables= $[q_0, q_1]$ and cartesian variable= $[x, y]$
Robots joint space to cartesian space transformation.
$x = L_1 * \cos(q_0) + L_2 * \cos(q_0 + q_1)$
$y = L_1 * \sin(q_0) + L_2 * \sin(q_0 + q_1)$
Cartesian space to joint space:
distance, $r = \sqrt{x^2 + y^2}$
$q_0 = arctan(y, x) - \frac{arccos({L_1}^2 + {L_2}^2 - r^2)}{2 * L_1 * L_2}$

$q_1 = 180 - \frac{arccos({L_1}^2 + r^2 - {L_2}^2)}{2 * L_1 * r}$

C. Workspace and coordinate transformation

The two-DOF arm has a non-uniform workspace and so we choose a rectangular area as the safe workspace as shown in Figure 3

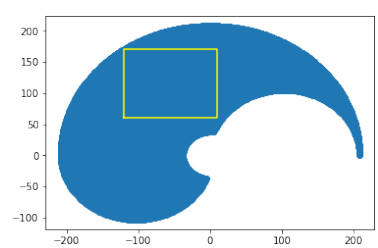

Figure 3. A rectangular safe workspace was chosen as the drawing area

Image space to robot's safe workspace coordinate transformation:

Robot workspace rectangle=
$rx_{min} = 0, ry_{min} = 0,$
$rwidth = 110mm, rheight = 100mm.$ Let, Image space rectangle = x0,y0, img_width, img_height
So, Image space coordinate (x,y) can be converted to robot space coordinate (rx,ry) as follows,
$rx = rx_{min} + (x - x_0) * \frac{rwidth}{img_width}$
$ry = ry_{min} + (y - y_0) * \frac{rheight}{img_height}$

IV. SHAPE DETECTION AND TRAJECTORY GENERATION

Convolutional Neural Network (CNN) based model was used to recognize 100 classes of shapes as known shape and collected shape paths data from adult writing was used to draw them. Kiddo learns new shapes from children demonstration as unknown shape. The whole process is described in the following sections.

A. Data collection

Analyzing the textbook, we have seen that up to the second grade, alphabet, numbers, and some geometric figures are taught along with some drawings. So we decided to teach the children this content with robots. We collected total 170752 image data of following shapes. All the shapes are described below:

Bengali Vowel (11): অ, আ, ই, ঈ, উ, ঊ, ঋ, এ, ঐ, ও, ঔ
Bengali Consonant (39): ক, খ, গ, ঘ, ঙ, চ, ছ, জ, ঝ, ঞ, ট, ঠ, ড, ঢ, ণ, ত, থ, দ, ধ, ন, প, ফ, ব, ভ, ম, য, র, ল, শ, ষ, স, হ, ড়, ঢ়, য়, ৎ, ং, ঃ, ঁ
Bengali digit (10): ০, ১, ২, ৩, ৪, ৫, ৬, ৭, ৮, ৯
English digit (10): 0, 1, 2, 3, 4, 5, 6, 7, 8, 9
English alphabet (26): A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z
Basic geometry (4): Circle, Square, Rectangle and Triangle

B. Training

The Convolutional Neural Network (CNN) [13] is used to train the images. Among the 170752 shapes, 136600 shapes were used in the training set and 34152 shapes were used in testing. Primarily, the testing accuracy of this dataset is good. However, in the time of experimenting with live camera feed taken from the whiteboard in real-time, the recognition results were not as good as found with the datasets due to light variance. For that reason, we divided the whole dataset into several sessions, and then the system gave accurate predictions. The system is arranged in such a way that the child can learn one shape at a time. For example, the children can learn Bengali consonant or a geometric shape at a time. This training is divided into 5 sessions. Table I shows 4 trained session. Another session is for drawing simple shape.

Table I
Content of session, training and testing data of each session and their accuracy

Learning item	Training data for each item	Testing data for each item	Testing accuracy in percentage
Bengali consonant	61733	15434	88
Bengali vowels and digits	33224	8307	93
English digits and geometry	21623	5406	96
English alphabet	20020	5005	96

C. known shape detection and saving paths

The known shape of four categories and their training information is given in table I. There is a total of 100 shapes. To make the robot's drawing looks natural we didn't choose standard font trajectories but human writing. A web app was developed which display a canvas on any touchscreen display and a user can write any shape using a stylus. We have collected shape data of those shapes from adults using the app.

D. Unknown shape to trajectory generation

Children often come up with innovative ideas and in case of drawing they love to show their drawing to others and ask if they can draw this too. In this section, we describe how the robot can generate trajectories from children's demonstrations so that the robot also draw the shape like the demonstrator. The following subsections describe the process.

1) Image thinning: Using the whiteboard marker makes shapes thick and for smoothing, "skeletonize" library from "skimage.morphology" is used to extract the skeleton of the shape [14].

2) RGB image to Grayscale image: RGB image is converted to grayscale black and white images. This grayscale image is then passed through a threshold filter and converted to a binary array where 0 contains the path of the shape.

3) Find Path trajectory: To find out the paths of an image, we have used the Depth-first Search (DFS) algorithm. DFS is a searching algorithm used to find paths between nodes, many searching problems can be solved by DFS such as maze solving [15]. An image is a binary matrix and here DFS starts from the first pixels and recursively check the eight nearest corners of the current pixels. This way, it traces all the paths in the image and generate x, y Cartesian coordinates of the pixels. These paths containing (x,y) values are saved in a file. DFS finds how many distinct path are in one shape. Suppose, a shape contains 4 paths then the robot pick up the pen and draw the shape by four times. How many times the robot picks up the pen depends on how many paths exist in the shape. Figure 4 shows the transformation of an

image. Figure 5 describe a flow chart of how the robot draws a shape by analyzing an image.

Figure 4. An image of the shape "home" is feed to the system (left), it generate a skeleton image(middle) and then segments four paths(right) marked as 1,2,3,4 in four colors. For each path, the robot lift-up the pen and move to the first coordinate of the path and draw it at once.

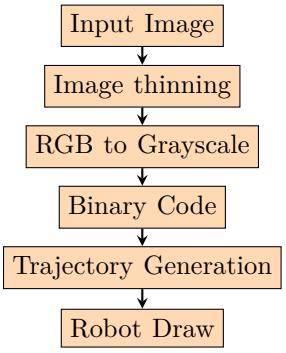

Figure 5. This flowchart describe the whole procedure of Image path to trajectory generation

E. Speech Conversation

We have used Google Speech recognition for Bengali language and Google Text to speech service on android platform. A simple reflex based conversation system was developed so that Kiddo can verbally interact with children.

V. EXPERIMENT

To ensure health safety in the current COVID-19 pandemic situation we have designed the experiment in two phases. In the first phase, children completed interaction with the simulated robot, and then in the second phase, the physical robot situated remotely mimic the simulated robot using the same data. In the first phase setup, a whiteboard was set on a desk, a camera over the board to capture what children write which was connected to a laptop placed beside the white-board where the simulated robot was running in the VREP simulator. An android phone with the same app as in the real robot was placed beside the white-board which was responsible for verbal communication with the children. In this setup, at a time one child can interact with the robot. Initially, they asked the robot "Can you write" the robot said, yes. Then they asked "What can you write", robot replied that it can write alphabets, digits, and geometry. Again, they wrote a letter on the board and asked "What I wrote", then the robot tried to recognize the shape. If the shape was known Kiddo said the shape name out loud otherwise it says the shape was unknown. The child instructs the robot saying "write", then if the shape is a known shape then Kiddo load the shape data and writes it otherwise it generates path trajectories from the demonstrated image and then draw it with its arm. In this way, children interact with Kiddo for a long time. In the second part setup, we placed the real robot on a desk and the interaction data from the simulated robot was used to draw those letters and shapes. Figure 6 shows the setup with the simulated robot (left) and the real robot (right).

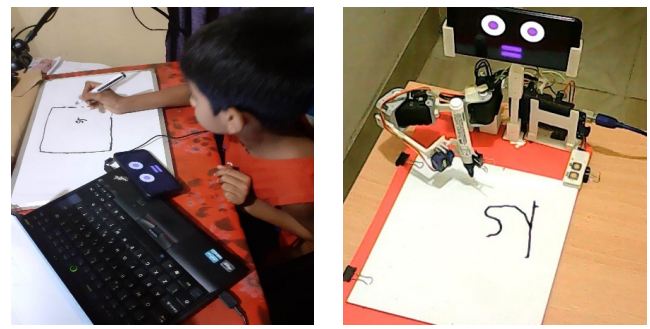

Figure 6. A child sitting in a chair table environment while interacting with simulated kiddo (left) wrote a Bengali consonant গ and the real robot shows a similar response as the simulated one (right). Due to current COVID-19 pandemic experiment with the real robot was done separately.

VI. RESULTS

Three children of average 6.3 years old were engaged with Kiddo in a total of 39 minutes and they wrote eighteen known shapes and taught Kiddo four new shapes during the experiment. On average each child exchanged writing activity with Kiddo for 13 minutes. Table II summarize the interaction.

Table II
Three children wrote, Bengali vowel, Bengali consonant, Bengali digits and some drawing. Each row describe distinct interaction

Child	Age	Known Shape	Unknown Shape	Interaction time in minutes
Child 1	6	অ, আ, ই, ০, ১, ২, ৩, ৪, ৫, ৬	no unknown shape	17
Child 2	8	ক, খ, গ, ঘ	Tree Bird	13
Child 3	5	A, B, C, D	Flower, Home	9

The robot was also able to segment paths from the images captured from child demonstration and drew them by holding a marker and moving its hand over the whiteboard similar to human handwriting. In the interaction, at first, the children were little inhibited to see how Kiddo was talking because the robot was running as a simulation inside a laptop and the voice was coming from an android phone which they knew as a distinct device.

After they were shown pictures of the real robot which was located in a different location due to COVID-19 pandemic and briefed how the laptop and the android animatronic display work as the robot they started showing friendly interaction with it. Figure 7 and Figure 8 show two known and two unknown shapes from the interaction with the children.

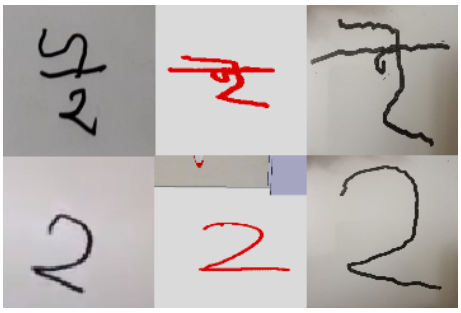

Figure 7. Children demonstrated Bengali vowel ই (top-left) and English digit 2 (bottom-left) and asked the robot to draw the same shapes. Robot draws in simulation (middle) and the real robot also wrote the shape (right)

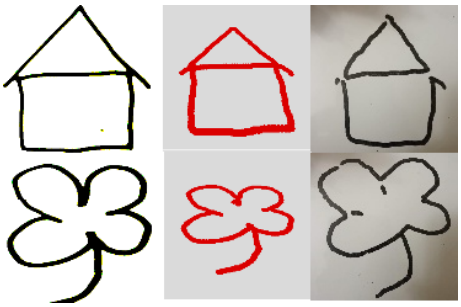

Figure 8. Kiddo learns new shapes from children demonstration. A child shows how to draw "home" (top-left) and another child showed how to draw a "flower" (bottom-left). Our system generated distinct paths from the images and then drew them in simulation (middle) and in the shared whiteboard by the real robot (right)

VII. CONCLUSIONS AND FUTURE WORKS

We have presented a new robot named Kiddo which was carefully designed to be an educational peer for children where Kiddo sits on a desk and its appearance in a standard chair-table study environment makes it look like an intelligent toy that gives a playful stimulation. From the interaction with the children grade 2 level knowledge of recognizing and drawing shapes was verified as the robot was able to recognize children drawing while they asked the robot to recognize them. The robot was successfully drawn 17 of 18 known shapes children demoed. Our third contribution of learning new shapes from the demonstration was also verified as Kiddo was able to segment paths from the images and draw them on both the simulation and real whiteboard with its hand. The conversation was limited which can be improved so that children can talk freely. As Kiddo was able to write and draw shapes in those interaction with the children, we conclude that it can be effective in early childhood learning in an enjoyable way. In future we will improve the conversation knowledge and develop interactive drawing games to make interaction more engaging.

VIII. ACKNOWLEDGEMENTS

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